DSA Day 27: Red-Black Trees

Book: Data Structures Through C in Depth - Ch6 §6.15 · pp 258-277 | Code in Python

Overview

RB trees fix violations after insert/delete using recoloring and rotations, keeping height ~2 log n.

Key concepts to master

- Insert then fix-up
- Uncle red -> recolor
- Uncle black -> rotate

Python implementation

```
# insert as red, then fix double-red via uncle's color
```

Complexity

O(log n).

Common pitfalls

- Forgetting to recolor the root black
- Missing an uncle case

Practice

- Insert a few keys, check no red-red
- Compare AVL vs RB insert cost